Optically Tunable Terahertz Metasurfaces on Fe-Doped β-Ga_2O_3

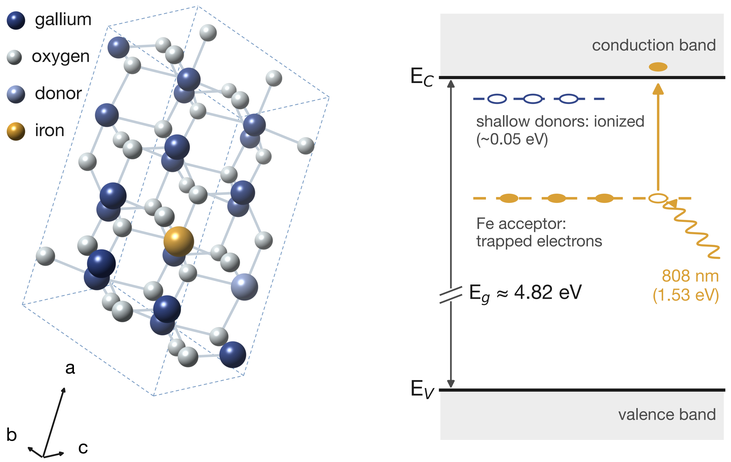
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Motivation

- THz systems need components that **reconfigure in operation**; electrical tuning is fixed at fabrication.
- **Optical pumping needs no contacts:** light tunes the substrate itself, via deep Fe levels in β-Ga_2O_3.

Material and Mechanism



- Fe traps the residual donors' electrons: **semi-insulating**, low-loss at THz.
- **808 nm light (1.53 eV)** releases them from the ~0.8 eV Fe level — **~25 S/m at 2 W** [1].

Measured Photoconductivity

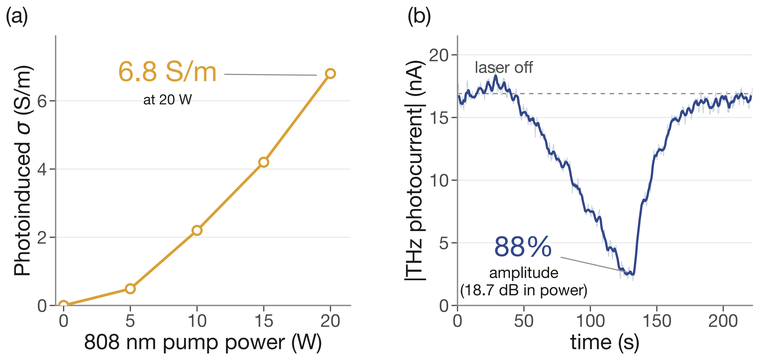


Figure 1. Measured. 90% recovery in 37 s.

- **The pump switches a real THz beam by 88%.** Bare crystal — no metasurface yet.
- σ is below [1]: different [Fe], donors, extraction.

The Device: H-SRR Metasurface

- **H-shaped split-ring resonators (H-SRR) over a ground plane**: THz is **reflected or absorbed**; the pump sets σ.

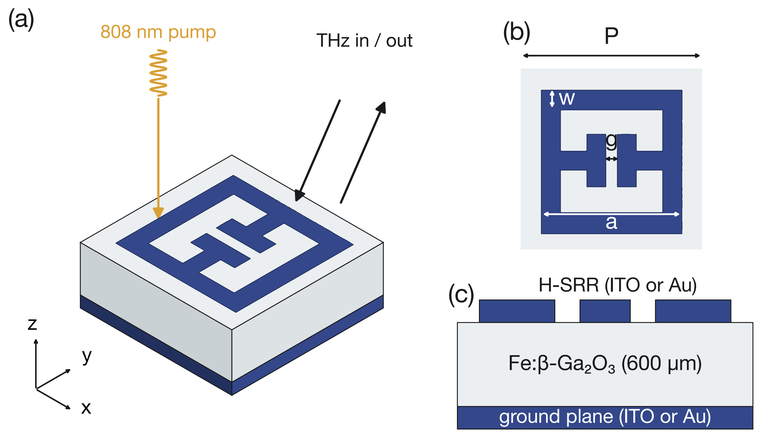


Figure 2. Unit cell; P = 150, a = 120, w = 15, g = 5 μm.

- **Gold prototype → all-ITO device:** ITO (0.5 μm) is transparent at 808 nm.
- σ(ITO) is **our own film**: 17 Ω/sq at 156 nm (450 °C anneal) → **3.7 × 10^5 S/m**.

ANSYS HFSS · Floquet ports · 20-pass adaptive mesh · σ(Ga_2O_3) 0–25 S/m.

Proof of Concept in Gold

- Critically coupled in the dark: **95.8% absorbed**. Photoinduced loss breaks this — **17.0%** at 25 S/m.

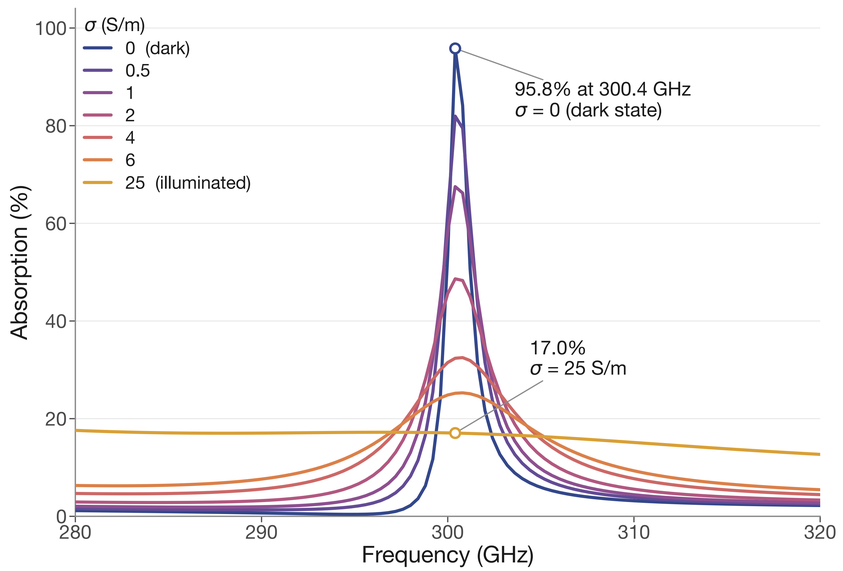


Figure 3. Simulated absorption, σ = 0–25 S/m (Au 0.2 μm on 724 μm Fe:β-Ga_2O_3).

82.2%	**95.0%**
absorption modulation	reflected-power modulation

TE / TM Switching

- The bare crystal switches broadband; **the metasurface adds selectivity** — chosen frequencies, one polarization.
- H-SRR asymmetry and (010) birefringence split TE from TM. **Four channels switch ≥ 10 dB.**

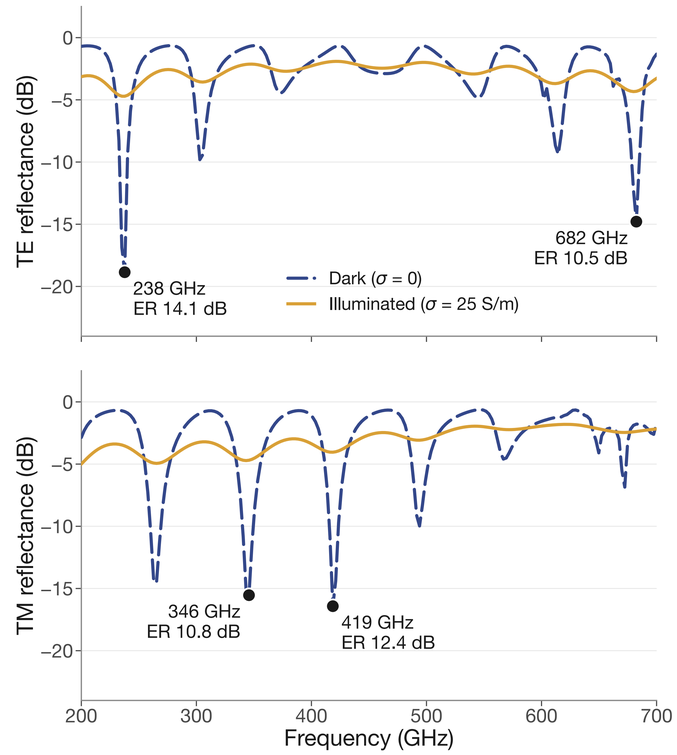


Figure 4. Simulated; repeating dips are Fabry–Pérot orders.

Channel	f_0 (GHz)	IL (dB)	ER (dB)
TE	237.7	4.7	**14.1**
TE	682.4	4.3	**10.5**
TM	345.7	4.7	**10.8**
TM	418.6	4.0	**12.4**

ER = dark-to-illuminated contrast; IL = loss when illuminated.

Conclusions and Next Steps

- Deep Fe levels in a UWBG host: **THz tuning without contacts** — measured **88% switching** at 463 GHz.
- **Simulated:** 82.2% absorption modulation (gold); four TE/TM channels ≥ 10 dB (ITO).
- **Next:** pattern the ITO metasurface, measure TE / TM across 200–700 GHz. Masks are designed.

References

[1] Jiang *et al.*, *Photonics* **9**, 233 (2022).

Acknowledgments

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